

60-GHz Low-Noise VGA and Interpolation-Based Gain Cell in a 40-nm CMOS Technology

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Abstract—This paper presents the design and measurement of an interpolation-based low noise and variable gain cell (IBA-cell) in the 60-GHz band, using a 40-nm CMOS technology. The interpolation-based gain cell is designed for an innovative analog beamforming front end, where the array pattern is not only controlled in the phase domain, but also wins the flexibility in the magnitude domain. The circuit specifications are first derived for the application at 60 GHz. Techniques to combine low noise figure (NF) with variable gain tuning are presented focusing on the NF and linearity (IIP₃) on the example of a 60-GHz low noise amplifier and variable gain amplifier. Subsequently, the design and measurement of the whole gain cell (IBA-cell) integrated into a single chip are reported with the technique of a cross-coupled feedback loop to reduce the phase variations over the gain tuning states and enhance the variations of IIP₃. The IBA-cell achieves 15.8-dB maximum gain and 6.5-dB NF at 57 GHz with the gain tuning range from −2 to 15.8 dB and IIP₃ varying from −11.3 to −16 dBm over the gain control range. The IBA-cell consumes a dc power of maximum 54 mW from 1.1 V.

Index Terms—Beamforming, CMOS, 5G, low noise amplifier (LNA), millimeter wave (mm-wave), phased array, receiver, 60 GHz, variable gain amplifier (VGA).

I. INTRODUCTION

SILICON-BASED phased array design at millimeter-wave (mm-wave) frequencies has been widely developed for decades [1]–[7]. Both the techniques of mm-wave MMIC [7]–[10] and phased array transceivers [1], [2] demonstrate promises for sustainable wide bandwidths to support multi-Gb/s data rate wireless communication and coherent gain benefits of beamforming to enhance the signal-to-noise ratio (SNR) of receivers and the efficiency of transmitters [11]–[14].

The mm-wave beamforming transceivers call for the innovation of architectures not only to increase the data rates and the communication ranges but also to reduce the cost, such as power consumption and chip area. Reference [15] presents a new beamforming array architecture [see Fig. 1(a)], developed in the European Research Council Advanced grant project

BROWSE, which reduces the number of time/phase compensation blocks like phase shifters in a beamforming system and provides the ability of array pattern shaping via vector signal interpolation. Reference [13] develops a hybrid system of a concurrent beamforming MIMO for high multiplexing gain and null steering. Reference [12] demonstrates a dual-vector phased array receiver from 25 to 35 GHz. It moves the quadrature generation in an RF beamformer to a global location to reduce the cost of quadrature generations and achieve a scalable front end, referring to a dual-vector topology. On the one hand, [13] uses one phase shifter per array element and presents a narrow-band beamforming system from 64 to 67 GHz. Reference [12] is applicable for a relatively narrow-band system, but cannot extend to a wideband (50% fractional bandwidth) one. On the other hand, interpolation-based array (IBA) architecture of [15] enables a broadband system with only three phase shifters for a full 2-D array.

The new interpolation-based architecture requires low noise amplifiers (LNAs) and variable gain amplifiers (VGAs), referring as the IBA-cell, and the calculated specification [see Fig. 1(a)] will be further discussed in the following. The IBA gain cell helps to improve the multiplexing gain but requires high gain tuning range (TR) and low linearity variation over the gain tuning states, addressed in [15] and [16]. The stand-alone single-ended LNA and VGA test chips have been presented in [16]. This paper shows more detailed explanations of the techniques to achieve a tunable gain cell with low noise figure (NF), high gain TR, and high linearity with small deviation over the gain TR. The IBA-cell is the key subblock in the interpolation-based beamforming architecture, and this is the first time to publish and address the design considerations for a full integrated IBA-cell. It includes the extension of the LNA and VGA presented in [16] to differential circuits and presents the design and measurement results of the full IBA-cell. This includes the design of a broadband transformer-based balun and power splitter.

First, Section II reviews the interpolation-based beamforming architecture and gives the overview of the circuit specifications compared with a classic analog beamforming system. Section III explains the detailed analysis and design on the NF and linearity performance in 40-nm CMOS technology. In Section IV, three ICs are implemented, including LNA and VGA test chips for the verification of the design strategy and the full integration and measurement results of the IBA-cell. The design techniques for a transformer-based power splitter

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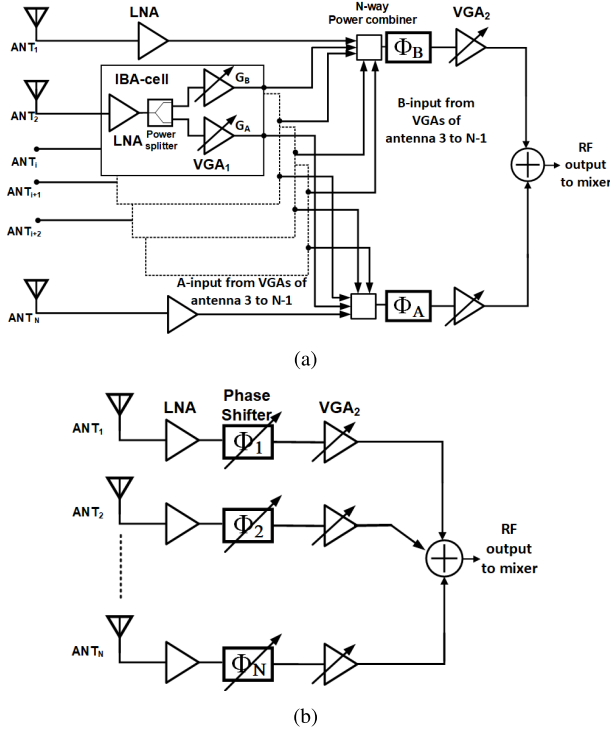


Fig. 1. System diagram of beamforming receiver front end. IBA architecture [15] versus a classic RF-phase-shifting one. (a) IBA RX front end. (b) Classic RX front end.

and balun are also included. Furthermore, a cross-coupled feedback loop phase compensation is presented for the IBA-cell. The system simulation is done using the measured results of the IBA-cell to prove the concept of system and discuss the circuit performance for this specific system. This paper compares the experimental results of the integrated IBA-cell and the test-chip low noise VGA (VGLNA) with the state-of-the-art work in the conclusions (see Section V).

II. IBA-CELL SPECIFICATIONS

1) *Principle of the IBA System:* For an RF-path analog phase shifting front-end receiver, shown in Fig. 2(a), the vector signals $\vec{A}$, $\vec{B}$, and $\vec{C}$ present the timed delayed signal received by each of the antennas of the array. The signal $\vec{A}$ is set as the reference and the signal $\vec{B}$ is from the most delayed antenna channel in one pico-cell of a linear array. So, the phase difference between $\vec{A}$ and $\vec{C}$ is Φ_C and that between $\vec{A}$ and $\vec{B}$ is Φ_B . There would be multiple $\vec{C}$ -like signals between $\vec{A}$ and $\vec{B}$. In a conventional beamforming architecture, each antenna channel deploys a time block (which can be implemented with a true-timed delay or a phase shifter, and the phase shifter is referred in this paper) to compensate the phase difference between adjacent channels before power combining. As presented in [15] for every signal $\vec{C}$, a decomposition is applied to reuse the phase shifts of the antenna channels $\vec{A}$ and $\vec{B}$, since

$$\vec{C} = a \times \vec{A} + b \times \vec{B}. \quad (1)$$

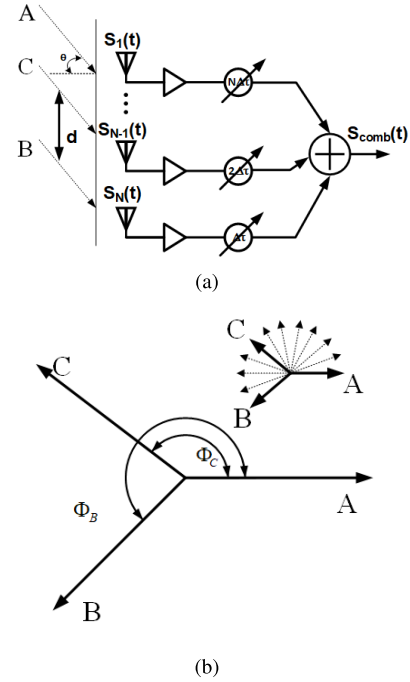


Fig. 2. Two base vectors A and B with a phase difference can be used to represent the vector C with respect to vector A . The inset shows that the vector C could be any of the N signals from the antenna elements of the array, in a linear array of RF-path phase shifting. (a) Principle of beamforming. (b) Vector diagram.

The coefficients (a, b) , referring to weighting factors, can be found by Euler's formula. More details can be found in [15]. It follows that the time delay corresponding to the phase shift required for vector $\vec{C}$ can be implemented through the time delays corresponding to the phase shifts Φ_A and Φ_B . An additional time delay corresponding to a phase shift Φ_C can be omitted. All signals from the various antenna channels will have a time delay with respect to the reference signals $\vec{A}$ and $\vec{B}$ and can be represented by such a superposition with individual coefficients (a_n, b_n) , where n represents the index of the antenna channels.

The system diagram of this new architecture is shown in Fig. 1(a). Compared with the conventional RF-path phase shifting architecture [see Fig. 1(b)], the main advantage is the reduced number of phase shifter elements. Commonly, the LNAs, phase shifters, and VGA₂s to compensate the magnitude mismatch between channels and power combiners of the array are presented. The main different block is the interpolation-based weighting block VGA₁ in Fig. 1(a). For both the IBA-based and conventional phased arrays, the weighting blocks are configured to adjust the shape of sidelobes and the main-to-sidelobe ratio. For an IBA array, the design considerations are presented in [15] in terms of a finite gain TR of the weighting VGA₁s in Fig. 1(a); 6-dB TR is validated to accommodate the system requirements of a linear uniform array, and a large gain (or attenuation) range of weighting blocks are adopted in the implementation of the IBA front-end receiver to eliminate the discontinuity of the array pattern aperture [15]. Furthermore, a Taylor weighting

TABLE I
CIRCUIT SPECIFICATIONS OF IBA RX FRONT-END

	Total	LNA	PwrSplitter	VGA ₁	2-level PwrCmb.	Phase shifter [19]	VGA ₂	1-level PwrCmb.
Gain (dB)	24	12	-1	8	-2	0	8	-1
NF (dB)	7.18	5	1	10	2	12	12	1
IIP ₃ (dBm)	3.6	-24.4	50	-16.4	50	0.4	-4.4	50
	Total	LNA	PwrSplitter	VGA ₁	2-level PwrCmb.	Phase shifter [20]	VGA ₂	1-level PwrCmb.
Gain (dB)	24	12	-1	8	-2	-15	21	-1
NF (dB)	8.96	5	1	10	2	15	12	1
IIP ₃ (dBm)	3.6	-24.4	50	-16.4	50	25	-19.4	50

TABLE II
CIRCUIT SPECIFICATIONS OF A CONVENTIONAL RX FRONT END
WITH REFLECT-TYPE PHASE SHIFTERS

	Total	LNA	RTPS [21]	VGA ₂	3-level PwrCmb.
Gain(dB)	24	12	-9	24	-3
NF(dB)	9.15	5	9	10	3
IIP ₃ (dBm)	10.6	-24.4	25	-21.4	50

TABLE III
CIRCUIT SPECIFICATIONS OF A CONVENTIONAL
RX FRONT END WITH STPSs

	Total	LNA	STPS [22]	VGA ₂	3-level PwrCmb.
Gain(dB)	24	12	-15	30	-3
NF(dB)	13.63	5	15	10	3
IIP ₃ (dBm)	8.6	-24.4	25	-27.4	50

can also be applied to the IBA system, for a low sidelobe ratio [17], [18], which requires for a high TR of weighting blocks as well. To overcome the tight NF required in the conventional architecture due to the passive or high insertion loss phase shifter following the LNAs directly, the active weighting blocks by means of a VGA are applied for the system. In all, higher power gain can be achieved close to antenna via VGA₁.

The active weighting blocks (VGA₁) relax the NF requirement of the LNA but lead to the penalty of high linearity of the succeeding blocks after VGA₁. As a result, the calculated specifications for the IBA-cell are included in Table I compared with a conventional phased array RF front-end with a reflective phase shifter and a switch-type phase shifter (STPS) in Tables II and III, respectively. The pico-cell of the array is assumed to have 4-antenna channels, and the maximum received power at the input of LNA is -35 dBm. We take the overall gain (G_{total}) of a single path as a reference for these three systems and compare the system NF and IIP₃. For blocks, the passive power splitter/combiner adopted is 3-dB equal power division or combining. Theoretically, the input third-order intermodulation (IIP₃) of passive power splitters

and combiners is infinitely large and assumed to be 50 dBm for calculation. In Table I, the gain of VGA₁ is 8 dB, with 2 dB margin to the requirement of interpolation weighting [15]. VGA₂ in Table I is set to be 8 dB for equalizing the gain mismatches between channels. As such, for a fair comparison, G_{total} equals to 24 dB for all system configurations. Therefore, VGA₂s in Tables II and III have a gain of 24 and 30 dB, respectively, by calculation; on the other hand, the VGA₂ can be deployed for Taylor weighting [18]. The input-referred NFs are calculated to be 7.18/8.96, 9.15, and 13.6 dB, respectively, in Tables I–III. The required IIP₃ of VGA₁ and VGA₂ in the IBA is higher than VGA₂s in the conventional architecture, but the gain is relaxed in the IBA; therefore, the cascaded IIP₃ in the IBA before RF downconversion blocks is lower than the other two systems.

From these numbers in Tables I–III, it becomes apparent that next to a good NF of the LNA, also the NF and linearity of VGA₁ are essential for an IBA-based array. Next to this, the requirements of VGA₂ in the IBA front-end and in the conventional one are fine gain tuning step-size and low phase variation [23]. Fine gain tuning step-size is to equalize the magnitude mismatches between adjacent channels, while low phase variation is important for large-sized beam-forming architecture and systems utilizing high-order complex modulation scheme, i.e., QAM-16. The state-of-the-art work on mm-wave VGAs has been reported in [24]–[26], but little attention is paid to the moderate IIP₃ degradation of linearity over the gain tuning. This is critical for not only VGA₁s in the IBA array but for VGA₂s in the conventional array, because the weighting value is independent of the received power. In the following, the design effort is put on the IBA cell, integrated with an LNA, a power splitter, and VGA₁, to achieve the performance of low NF, high gain TR, and low linearity/linearity variations (IIP₃).

III. DESIGN TECHNIQUES

The design considerations for the IBA cell are presented in this section via an LNA and a VGLNA, including the measurement results.

A. Low-Noise Design Topology

Fig. 3 shows the circuit schematic of the LNA. It consists of a two-stage cascode amplifier. Similar to the common-source

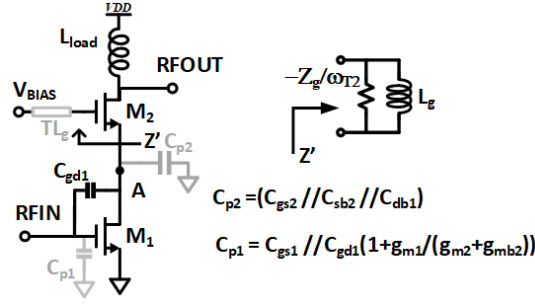


Fig. 3. Single-stage cascode amplifier with parasitics. (The biasing circuit is omitted.)

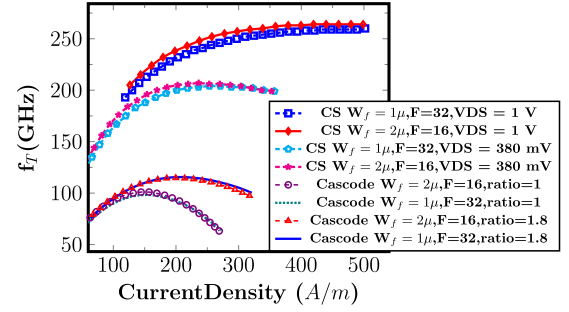
amplifier, the parasitic capacitance C_{gd} of M_1 has to be taken into consideration for the input impedance at high frequency. Seen from M_1 's gate, the Miller effect of C_{gd1} is suppressed by the transconductance g_{m2} of the cascoding transistor M_2 . So, at mm-wave frequencies, the cascode structure has advantages over the common-source one. However, it also introduces a pole at node A due to the shunt capacitances C_{gs2} , C_{sb2} , and C_{db1} . The most significant influence is the transformation of the parasitic at the gate of M_2 , at mm-wave frequencies. The inductive part L_g of TL_g is transformed into a shunt network of a scaled inductance and a negative resistance, which will be crucial for stability, as shown in Fig. 3.

According to the above-mentioned analysis, the two main design targets are: 1) noise optimization and 2) trade-off between the power gain margin and stability. In [27], the device sizing method as a function of NF performance is presented. The analysis in a 40-nm CMOS is carried out to properly size the transistors for the LNA design. Fig. 4(a) and (b) shows the general optimum design point in terms of speed and noise, in which the NF reaches its minimum value as configured as a common source structure with higher V_{ds} voltage. The NF degrades in the cascode structure from 1.8 to 3.6 dB. Meanwhile, the parasitic capacitances of the pads are in parallel with the source impedance and affect the noise and impedance-matching of the LNA. The input impedance of the LNA is designed to match the expected conjugate impedance ($27-j \times 17$) Ω seen from the probe taper instead of the 50- Ω reference available for the measurement. Note that the pad model used for RF-probing is EM-extracted, as an equivalent 45-fF lumped capacitance model. As a result, depicted in Fig. 5, the inductor L_G and L_{S1} are simultaneously designed for noise and power matching [28].

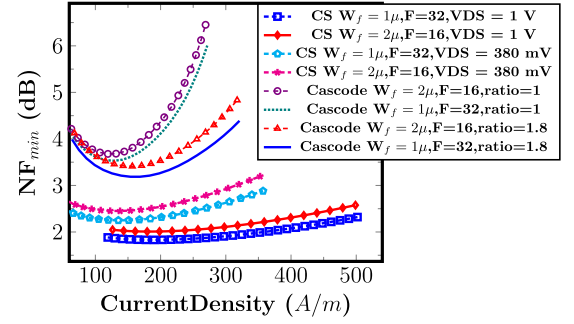
Another noise optimization technique in Fig. 5 is applied between the transistors M_1 and M_2 [29]. The equivalent circuit model of the cascoding configuration in Fig. 6 shows that the two main noise sources (the voltage noise source $\overline{e_{n2}^2}$ and the drain current source $\overline{i_{nd2}^2}$) of the cascoding transistor contribute to the noise at the output node, as modeled in (2b) [30], [31]

$$\frac{I_{n,out}}{I_{nd2}} = \frac{\frac{1}{(g_{m2} + \frac{1}{r_{ds2}})}}{Z_A + \frac{1}{(g_{m2} + \frac{1}{r_{ds2}})}} = \frac{1}{1 + Z_A \times (g_{m2} + \frac{1}{r_{ds2}})} \quad (2a)$$

$$\frac{V_{n,out}}{V_{n2}} = \frac{-(Z_L \times g_{m2} \times r_{ds2})}{r_{ds2} + Z_L + Z_A(g_{m2} \times r_{ds2} + 1)} \quad (2b)$$



(a)



(b)

Fig. 4. (a) f_T and (b) NF_{min} as a function of the input transistor's current density in a 40-nm CMOS, with different transistor sizing, where W_f is the transistor width, F is the number of fingers, and ratio corresponding to the ratio of the width of cascoding transistor to that of common-source transistor in the cascode structure.

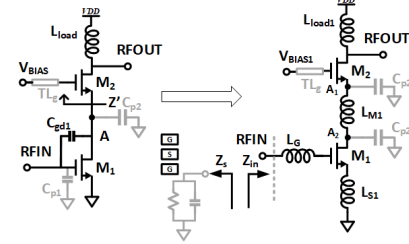


Fig. 5. Schematic of a single-stage cascode amplifier with a noise matching network. (The biasing circuit is omitted.)

where Z_A is the simplified impedance of the passive network at node A, which equals to $Z_A = (sL_M + (1/sC_{p2a})) \parallel (1/sC_{p2b})$ (see Fig. 5). Equation (2b) reveals that the higher the impedance of Z_A , the better the noise performance. Moreover, the inductor L_{M1} increases the equivalent output impedance at the drain of M_2 by $Z_A \times (g_{m2} \times r_{ds2})$ and the output impedance of M_1 seen at node A1, respectively. So, when L_{M1} is resonant with the parasitic capacitances of C_{p2a} and C_{p2b} at the frequency of interest, it leads to a high gain, a good NF, and a high isolation of the amplifier.

B. Variable Gain Topology

At mm-wave frequencies, there are three main techniques to implement gain tuning in amplifiers: 1) gain tuning by gate bias control [see Fig. 7(a)]; 2) current steering techniques [see Fig. 7(b)]; and 3) attenuators in the signal path [see Fig. 7(c)].

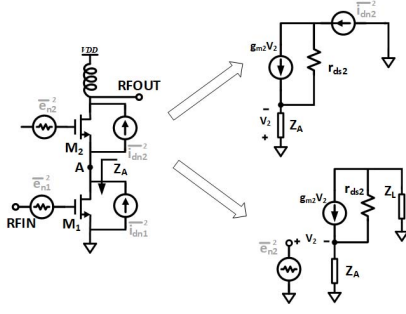


Fig. 6. Simplified equivalent circuit model analysis for the effect of noise of the cascoding transistor at the output RFOUT.

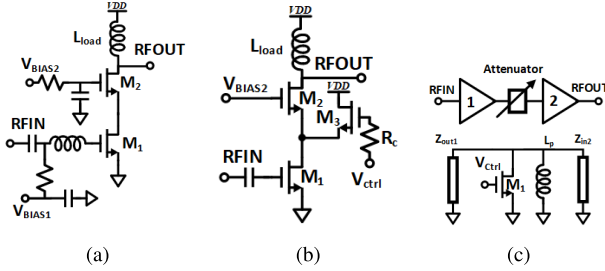


Fig. 7. Three main topologies of VGAs utilized at mm-wave frequencies. (a) Gate biasing. (b) Current steering. (c) Attenuator.

TABLE IV
VGAs COMPARISONS

	Gate biasing	Current-steering	Attenuator
Digital-Assist	—	++	++
ReturnLoss	—	++	—
IP1dB	—	—	++
NF	—	—	—
Implementation	++	—	—

For the IBA receiver front end, the gain TR and the high linearity are the key performance parameters of VGA_1 [see Fig. 1(a)]. Table IV summarizes the advantages and disadvantages of these three topologies.

The first one is gate biasing control, where for a cascode amplifier, the changes of common source gate biasing voltage will change the dc current and the transconductance g_m , so the conversion gain changes. Its implementation is easy but shows drawbacks in other parameters, i.e., the ability of digital assistance for the complex calibration scheme, input return loss, and NF and linearity, when compared to the other options. The second one is current steering. In contrast to the first method, the dc current of the g_m stage M_1 is kept constant, and the current splits into two paths: to the signal path via M_2 and to the current steering transistor M_3 . The advantage of the current steering technique is that the linearity is mainly determined by the common-source transistors, as shown in [32].

So, this topology has no enhancement on IP_{1dB} and less degradation of the NF. Also, it benefits from stable input/output return loss, making it popular at mm-wave frequencies. The third choice is a passive attenuation network. The attenuator is composed of the switch-on resistor of

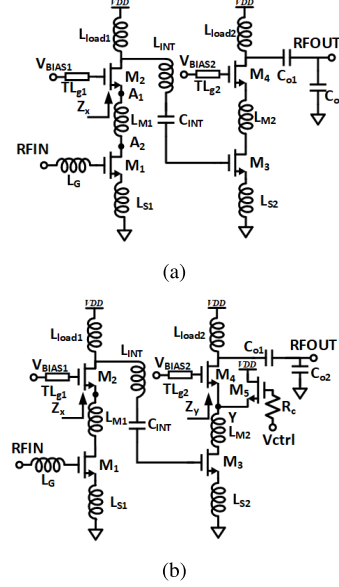


Fig. 8. Schematic of test chips. (a) Schematic of LNA. (b) Schematic of VGLNA.

transistor M_1 and the parasitic capacitances, together with the ac terminated inductor L_p . This technique has a wide variation in terms of input/output return loss; To relax this issue, a two-stage topology is mostly applied, and still, it will degrade the NF. As VGA_1 is the second block after the antenna in the receiver link, it contributes significantly to the overall NF and linearity. Minimizing the NF performance degradation is important and IP_{1dB} is traded-off by means of the dc power consumption.

C. Experimental Verification With LNA and VGLNA Test Chips

The full schematic of the LNA is shown in Fig. 8(a), as a two-stage cascode structure, with noise optimization techniques by means of inductive degeneration L_{s1} and L_{s2} and noise matching L_{m1} in a cascoding structure. Besides, the inductive degeneration L_{s2} helps to improve the linearity of the amplifier [33]. In the implementation, the short transmission line TL_{g1} at the gate of M_2 , with the equivalent inductance L_g , can not only be used as bias line but also boosts the gain of the cascode amplifier by introducing a negative resistance at node A [34], [35] of approximately $Z_{Aneg} = (-\omega^2 \times L_g \times C_{gs2}/g_{m2})$ (see Fig. 3). Furthermore, the transmission line is implemented in the three top metal layers in stack to increase the Q -factor, and the structure is shielded from the substrate by metal strips in the low-layer metal. Because TL_{g1} contributes an inductance value of L_g seen at node A as well, L_{m1} is tuned to compensate L_g into the resonance network at node A. To achieve a stable operation, g_{m2} of M_2 , the electrical length and the characteristic impedance of TL_{g1} at the gate of M_2 and L_{m1} have been optimized. This tradeoff is complicated because at node A, a negative resistance Z_{neg} , an inter-stage inductor L_{m1} , and several parasitic capacitors exist simultaneously. The same design technique is applied for TL_{g2} in the second stage.

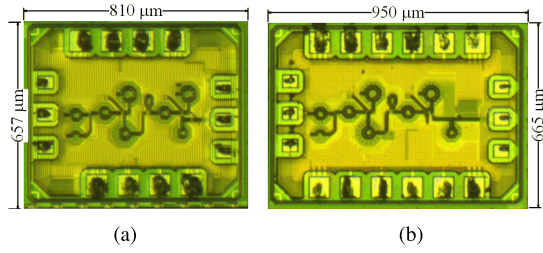


Fig. 9. Die photograph of the amplifiers. (a) Single-ended LNA. (b) Single-ended VGLNA.

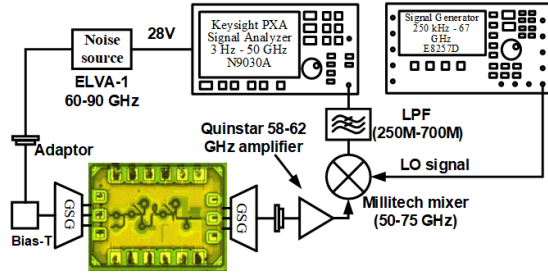


Fig. 10. Measurement setup of NF.

The structure we choose for the implementation of VGA_1 in the IBA-cell is the current steering VGLNA, as shown in Fig. 8(b). The VGLNA has two stages: the same design topology of the first one as for the low-noise amplifier [see Fig. 8(a)] and the second stage implements a gain tuning function. Depending on the difference between the voltages V_{ctrl} and V_{bias2} , the dc current in the output branch is steered between M_4 and M_5 . At the second stage, the inductor L_{M2} is placed before node Y, pushing the pole of the transfer function to high frequency to obtain a flatter gain at high frequency. For the stability, R_c is added to the gate of M_5 to compensate the negative resistance seen at node Y, taking g_m and C_{gs} of M_5 into consideration.

D. Layout and Measurement

The circuits have been fabricated in a 40-nm CMOS technology. The die photographs of the LNA and the VGLNA are shown in Fig. 9 with the area of $810 \mu\text{m} \times 657 \mu\text{m}$ and $950 \mu\text{m} \times 665 \mu\text{m}$, respectively [16]. The S-parameters are measured with a 4-port network analyzer PNA-X N5247A from Keysight, and the NF is measured in the setup shown in Fig. 10. A 60–90-GHz ELVA-1 noise source is used. Because the spectrum analyzer can only work up to 50 GHz in the NF measurement mode, frequency downconversion is implemented via a Millitech mixer that downconverts the upper sideband noise into the sub-1-GHz band. Moreover, a Quinstar 58–62-GHz high power gain amplifier acts as a preamplifier to compensate for the conversion loss (7–11-dB loss) after the mixer. This is especially relevant, as the DUT has a considerable low gain and low input power in the noise measurement.

1) *Measurement Results of LNA*: The measured S-parameter results are shown in Fig. 11 in comparison to the simulation results. The measured S_{21} is 6.7 dB and the 3-dB bandwidth is

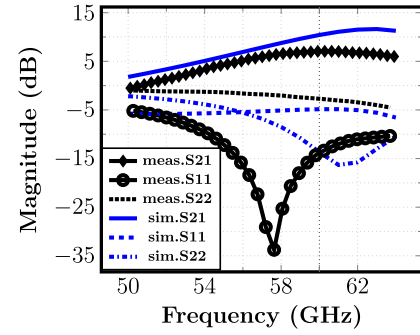


Fig. 11. Measured and simulation results of LNA in the band of 50 and 64 GHz.

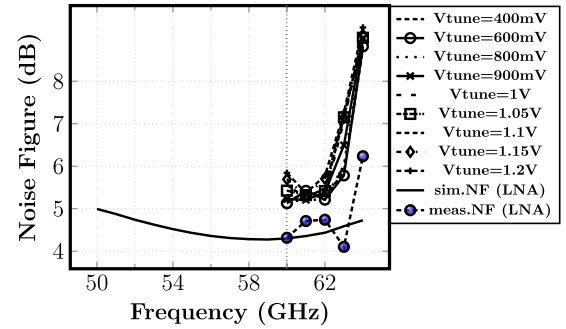


Fig. 12. Measured (60–64 GHz, solid line) and simulated (50–64 GHz, dashed line) NF of the LNA and the VGLNA over all the gain states.

9 GHz from 55 to 64 GHz. The measured S_{11} is below -10 dB from 54 to 64 GHz, where the S_{11} curve shows an additional minimum value at 58 GHz. Compared with the simulated results, the S_{21} curve is 4 dB lower than the simulated value in the band from 60 to 64 GHz. The deviation comes from the mismatch of the transmission-line-based inter-stage matching network in the implementation, as the floating-shielding structure is applied without the well-defined current-return ground for the transmission line; furthermore, the RC parasitic of the transistors is double extracted and overestimated in the design.

The NF measurement results are shown in Fig. 12 in the band from 60 to 64 GHz only, limited by the ELVA-1 noise source and the bandwidth of the Quinstar amplifier. It can be observed that the NF is 4.3 dB at 60 GHz close to the simulated results; at 64 GHz, the low gain of the Quinstar amplifier results in large measurement deviation from the simulated result. The IIP_3 is -12.8 dBm at 60 GHz with 50-MHz channel spacing. The dc current is 12 mA at 1.1-V supply voltage.

2) *Measurement Results of the VGLNA*: The VGLNA is measured with the same setups as the LNA. To verify the gain tuning, the tuning voltage V_{ctrl} is set from 0.4 to 1.2 V. The S-parameters with different gain tuning states [see Fig. 13(a)] show that the maximum S_{21} is 6.67 dB at 58 GHz with V_{ctrl} of 0.4 V. The measured S_{21} decreases as the tuning voltage increases. When tuning the control voltage V_{ctrl} , the tuning gain range is 6 dB with 15° phase variation, which can be improved by adding a compensation capacitor at the gate of the current-steering transistor M_5 with no extra power

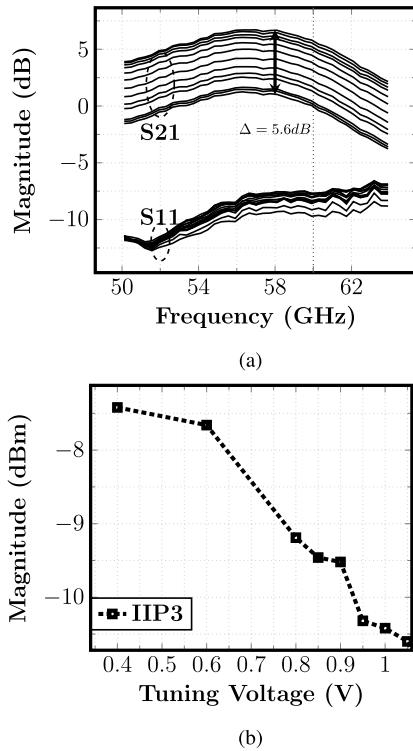


Fig. 13. Measured results of the VGLNA at all gain states. (a) Measured S_{21} and S_{11} . (b) Measured IIP₃ at 60 GHz with 50-MHz spacing.

consumption. The NF (see Fig. 12) varies from 5.13 to 5.84 dB at 60 GHz as V_{ctrl} increases. At the same time, the IIP₃ [see Fig. 13(b)] degrades from -7.42 to -11.33 dBm at 60 GHz. The input matching is not affected by the gain tuning. Compared to the performance of the LNA, the VGLNA has a slightly higher NF at the maximum gain state because of the noise contribution from the current-steering transistor which is not well suppressed by the first stage amplifier. It requires 11-mA dc current from 1.1-V supply voltage.

IV. IBA CELL DESIGN AND MEASUREMENT RESULTS

A. Differential LNA/VGA Circuit Topology

Major goals of this section include the subsystem integration of the IBA cell, as shown in Fig. 1(a), consisting of an LNA, a power splitter, and two VGAs, referred as IBA-cell, with the requirements of the minimization of the power consumption, a maximum gain above 10 dB, a high amplitude control range (maximum gain or maximum attenuation), and minimum NF. Moreover, the phase mismatch between two weighting factors (a, b) is critical to the beam pattern. The design and implementation are explained in detail.

As shown in Fig. 1(a), the output of the LNA is split equally and is input to identical A- and B-path VGA (VGA₁) as weighting function for the interpolation-based beam steering. The schematic of the IBA cell is shown in Fig. 14. The Gilbert-cell-based (M_{13} , M_{14} , M_{15} , and M_{16}) polarity switch is applied in the second stage of the weighting VGA₁ to get two in-phase or out-phase signals OUT_A and OUT_B according to the calculated values of (a, b), in which the switch control signal is SW . Moreover, to compensate for the linearity degradation

in low power state, the Gilbert-cell VGA is at the last stage of the IBA-cell [36]. The single-ended LNA, which has been presented in Section III, is sensitive to parasitic ground inductance between the ground return path of the RF signal source and the ac ground of the first stage particularly. The differential architecture can alleviate the problem of parasitic source degeneration. The pseudodifferential LNA stages are based on a cascode structure without inductive degeneration (see Fig. 14). A transformer is used at the LNA input as a balun. It can simultaneously provide input noise and power matching as well as ESD protection. As for a compact layout, transformers are further utilized for both interstage matching and loading purpose. For the design of VGA₁, the topology of current steering is applied for the IBA-cell.

B. Transformer-Based Balun

An impedance matching of a 1:2 transformer is implemented to simplify the simultaneous noise and power matching network and to achieve a compact chip size. Fig. 15 shows the 3-D view of the on-chip balun. It consists of 1 turn in the aluminum (Al) layer as the primary winding and 2 turns in M_8 and M_7 as the secondary winding. The line width in the Al layer is $15 \mu\text{m}$ with the slotted space of $3.5 \mu\text{m}$, and for M_7/M_8 , the line width and the space are 5 and $7 \mu\text{m}$, respectively. As shown in Fig. 15(c), the top metal layer is slotted to reduce the resistive loss [28]. In order to maximize the magnetic coupling between the primary and the secondary winding, broadside coupling is implemented as set by the distances R_1 and R_2 .

C. Transformer-Based Power Splitter

Another critical passive component of the IBA-cell is a power splitter (see Fig. 14). A transformer is applied here to couple two amplifiers and achieve a two-path efficient power splitting. The butterfly-shaped transformer [shown in Fig. 16(a)] is designed on the top two metal layers: M_8 and M_7 , with the center tap CT_p implemented in the Al layer for dc supply and CT_s in the lower metal layer for dc biasing of the succeeding stage amplifier. The primary winding in M_8 is the inductive load of the LNA, while the secondary winding in M_7 is duplicated as the two wings of the “butterfly,” which is utilized as the impedance transformation and input network of two VGA₁. The current direction in these two secondary windings is orthogonal to enhance the isolation between the two output ports, as shown in Fig. 16(b).

The EM simulation is done for differential input/output impedance ($Z_{\text{diff}} = 100 \Omega$) configuration. The simulated results are shown in Fig. 17. The passive loss of the power splitter is around 1 dB of which -3.82 dB for the A-path and -4.04 dB for the B-path; the phase mismatch between the two paths is less than 1° . Fig. 17(b) shows the return loss and the isolation. The 3-dB bandwidth of the return loss is from 52 to 76 GHz, while the isolation is below -20 dB in the band.

D. Phase Variation Compensation Technique

The phase variation shown in the single-ended topology is not good enough for the array pattern and the compensation

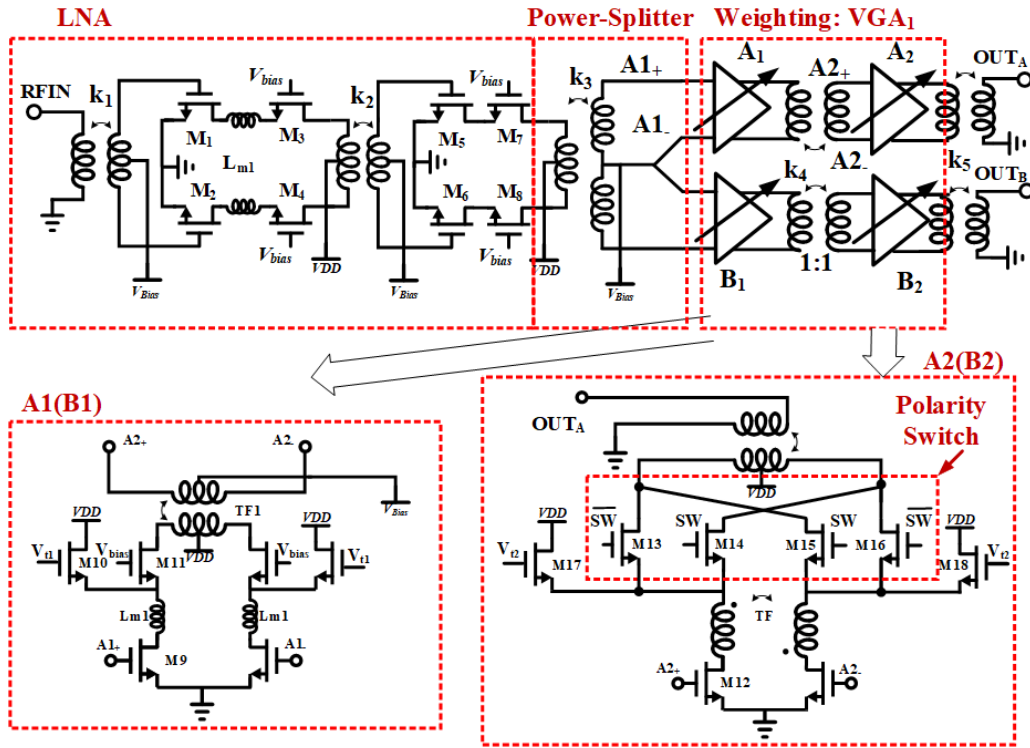


Fig. 14. Schematic of IBA-cell (top), with the differential structure of LNA (top left) and VGA₁ (top right). Bottom left: first stage. Bottom right: second stage.

capacitor does not suit for a differential circuit because it requires a well-defined ground connection. Therefore, another phase compensation technique is exploited in this section.

In [18], it is pointed that in a current-steering cascode amplifier, there are two main causes for the phase variation: one is the high-frequency response of the cascoding transistor leading to the poles/zeros shift and the other is the change of the equivalent capacitance seen by the cascoding transistors' drain. To compensate, two optimization techniques are applied and the schematic of A1(B1) amplifier in Fig. 14 is modified accordingly. The new schematic is shown in Fig. 18(a) with: 1) cross-coupled feedback c_f and 2) peaking transformer TF_m .

The feedback loop is voltage-current feedback by cross coupling between the drain node and the source node in the common gate topology. The simplified small signal model is shown in Fig. 18(b), and the extra benefit of the feedback loop is that it enhances IIP₃ [37]. The peaking transformer of TF_m has a self-inductance of 130 pH and a magnetic coupling factor of 0.4. Comparing to a set of two peaking inductors L_{m1} in Fig. 14, the symmetric layout of transformer does not need a ground connection and isolation for each peaking inductor. In all, the relation of the feedback capacitor c_f to the phase variation is shown in Fig. 18(c) that there is an optimal value 12 fF, in which the phase variation is as small as 1.6°.

For such a small capacitor c_f , careful layout and modeling are required [22]. Therefore, a Gilbert-cell switch in CMOS technology is utilized to play the equivalent role of feedback loop c_f , as shown in Fig. 19(a). As known, when the switch is OFF, the drain and the source are connected through c_{bd}

and the parasitic capacitance equals to $(c_{gs} + c_{gd})$ from the small signal circuit model of transistors [see Fig. 18(b)]. The simulation results plotted in Fig. 19(b) show that the optimized phase compensation frequency is at 60 GHz. The phase variations are 1.5° at 60 GHz, 1.6° at 64 GHz, and 6° at 57 GHz, while they are 24°, 25°, and 35°, respectively, without compensation. To verify, the phase compensation technique is applied to the A2(B2) stage as in Fig. 14, while not to the A1(B1) stage for comparison.

E. Layout and Measurement

The circuit has been fabricated in a digital 40-nm CMOS technology. The die photograph of the IBA cell is shown in Fig. 20, where the chip is dc wired bonded to a PCB. It occupies an area of 1735 $\mu\text{m} \times 925 \mu\text{m}$, including all pads. The same measurement setup is applied, as shown in Fig. 10. The circuit drives 54 mA from the dc supply voltage of 1.1 V.

The comparison of the simulated and measured S-parameter results is shown in Fig. 21. The simulation results show that at the maximum gain state, the input bandwidth is from 45 to 65 GHz. Meanwhile, the simulated S_{21} is 18 dB at 60 GHz. The simulated S_{22} is well matched beyond 60 GHz. On the other hand, the measured results indicate that, when polarity switch signal SW is "1," the results are subscripted with "P," while SW of "0," they are subscripted with "N." In general, the measured S-parameter performance shifts to the lower frequency, and for both SWs of "0" and "1," the maximum gain frequency shifts from 60 to 57 GHz with the value of 15.68 dB.

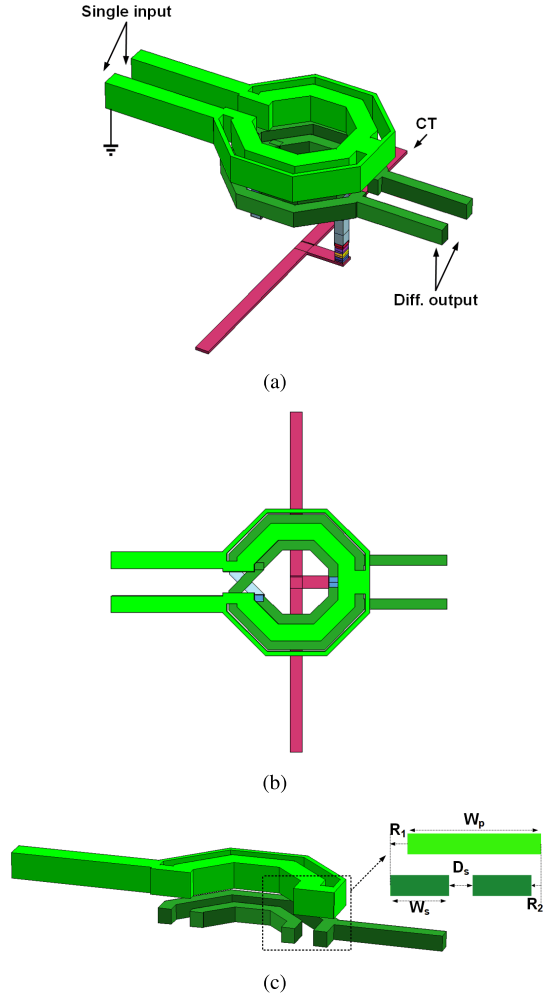


Fig. 15. EM model of the input balun. Dark green: M_8 . Blue: M_7 . (a) 3-D view. (b) Top view. (c) Cross-sectional view.

1) *Mismatches Between Two Paths*: Another important parameter for the IBA-cell is the mismatch between the A-path and the B-path weighting VGA_1 , which is evaluated and measured in Figs. 22 and 23. The phase difference of S_{21} for SW = “0” and SW = “1” is 184.5° for A-path VGA_1 and 173.5° for B-path VGA_1 . Within A-path or B-path, the S_{21} matches each other in the band of 54 and 64 GHz but has the deviation of 7 dB at 50 GHz for SW = “0” and SW = “1.” This is because the implementation of the isolation between the two output ports of the power splitter is not as good as simulated. Moreover, the layout of the two-path VGA_1 section is close to each other and introduces coupling between them. The isolation between the two output ports of the power splitter cannot be measured directly but is estimated by means of comparison of the phase mismatches, as shown in Fig. 24. The measurement is configured with all the gain states set for the maximum gain, while the SW for both A-path and B-path VGA_1 is set to be “1” and “0.” Since the overall 4-state phase response is measured, the phase mismatch of 8° and 15° is observed at 60 GHz for SW = “1” and SW = “0,” respectively.

2) *Noise Figure Measurements*: The NF is measured for the IBA-cell with the same setup, as shown in Fig. 10.

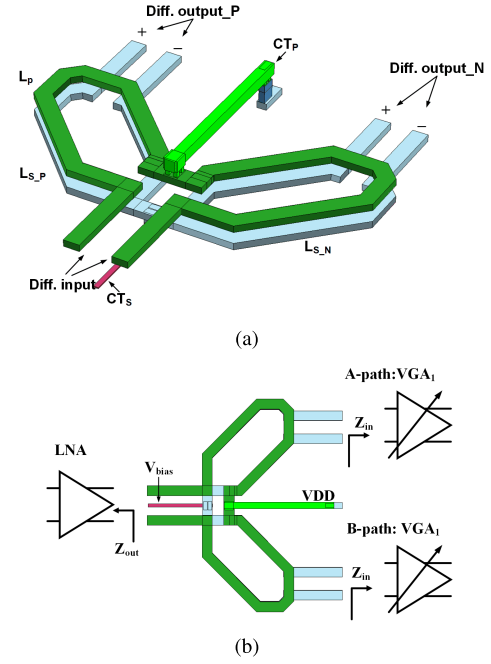


Fig. 16. EM model of a 3-dB power splitter. Dark green: M_8 and Blue: M_7 . (a) 3-D model. (b) Top view with amplifiers.

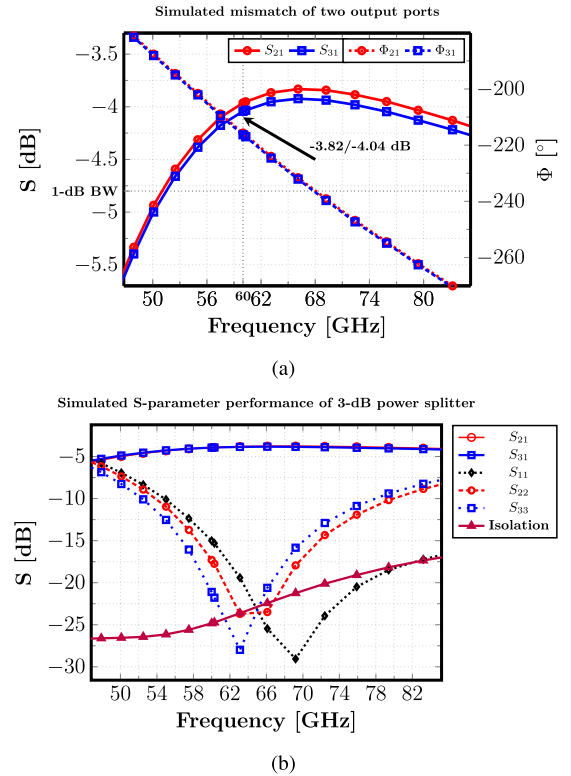


Fig. 17. EM simulated results of transformer-based power splitter. (a) Mismatch of two output ports. (b) S-parameters.

The measured results are shown in Fig. 25 compared with the simulated results. The calibration is done so that the gain measured by the spectrum analyzer is almost matched with the gain measured by the VNA. The NF is measured

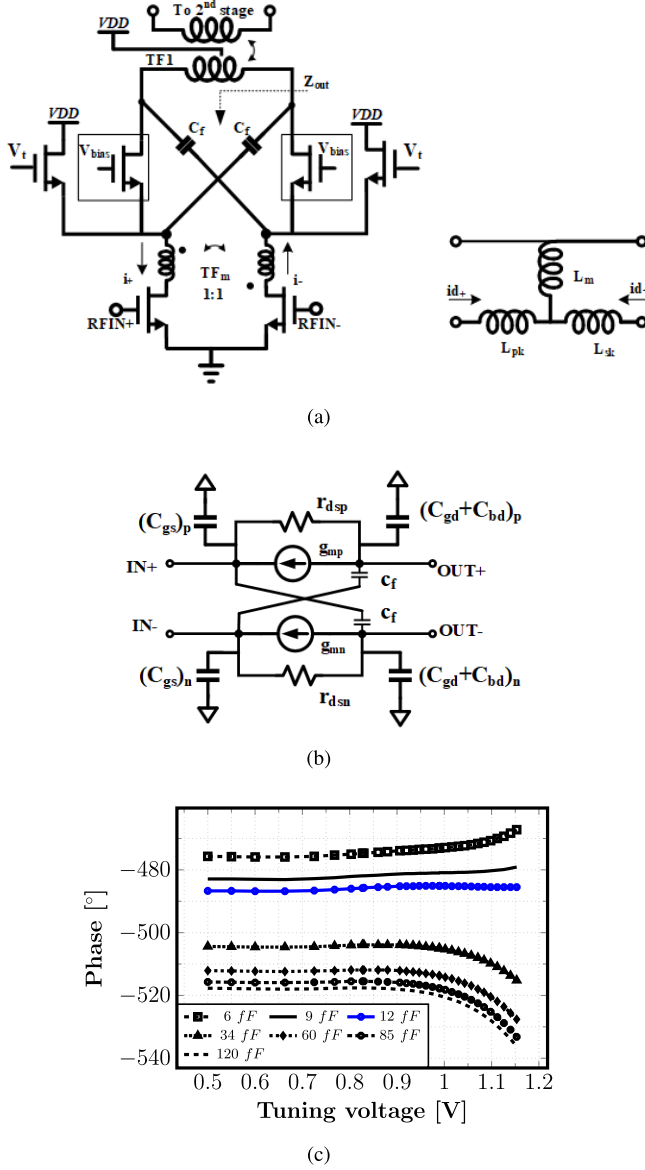


Fig. 18. Phase compensation technique. (a) Schematic of optimized first-stage VGA utilizing cross-coupled feedback loop. (b) Simplified small signal model of cross-coupled feedback loop. (c) Simulated phase variations over tuning states for different feedback capacitors c_f .

in the band of 58 and 64 GHz, and the excess noise ratio values of 58 and 59 GHz are set the same value as that at 60 GHz, which are not well characterized by the manufacture and introduced measurement error. The measured minimum NF at 61 GHz is 6.5 dB, while the simulated NF is 5.5 dB. By the limited working bandwidth of the external amplifier (58 to 62 GHz), the NF measured at 63 and 64 GHz also has some measurement errors. Overall, the trend of the measured NF matches the simulated one.

3) *Linearity IIP₃ Measurements*: Figs. 26 and 27 show the linearity of the IBA cell. At 60 GHz, with the maximum gain state setup, the input 1-dB compression point is -23 dBm, as shown in Fig. 26. Moreover, at 60 GHz, the tuning voltage of V_{t1} and V_{t2} changes simultaneously from 0.4 to 1.2 V to measure IIP₃, and the IIP₃ deviations are shown in Fig. 27.

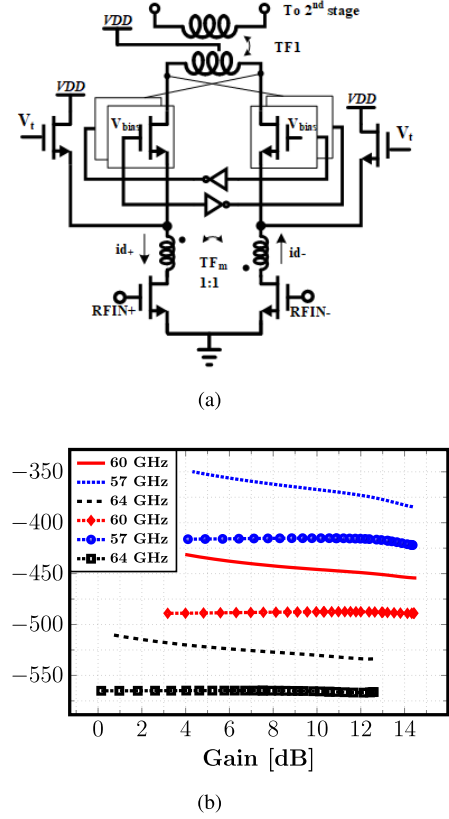


Fig. 19. Modified Gilbert-cell-based A1(B1) stage as in Fig. 14 with the simulation result comparison. (a) Schematic of optimized first-stage VGA utilizing Gilbert cell. (b) Optimized simulation results (in lines with markers) comparing to those without compensation (in lines without markers).

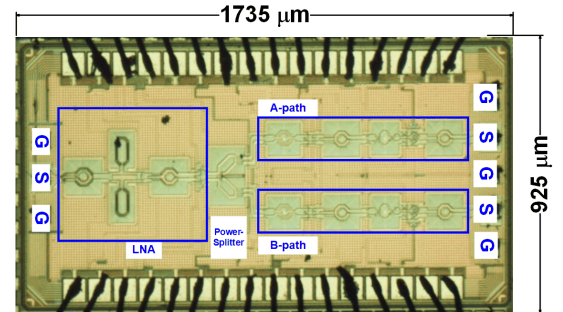


Fig. 20. Die photograph of an IBA cell implemented in a 40-nm CMOS technology.

At the maximum gain state, the IIP₃ achieves the best value of -11.3 dBm and varies from -11.3 to -16 dBm. The deviations between three IC samples have been checked as well, and the IIP₃ for the three chips is in the range from -11.3 to -16 dBm. The linearity of the amplifier is reduced at the low gain mode because of the reduction of the dc current of cascoding transistors.

4) *Tuning Measurements*: The last critical performance to verify by measurements is the gain TR. Fig. 28 validates that the return loss of the IBA-cell has quite small variations, which suits for the high-level integration. The plotted S_{21} curves are the measured tuning results of the A-path VGA₁ with

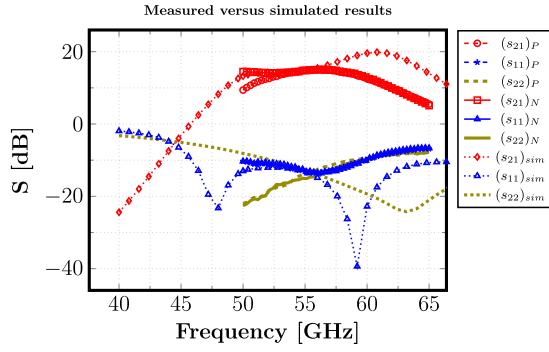
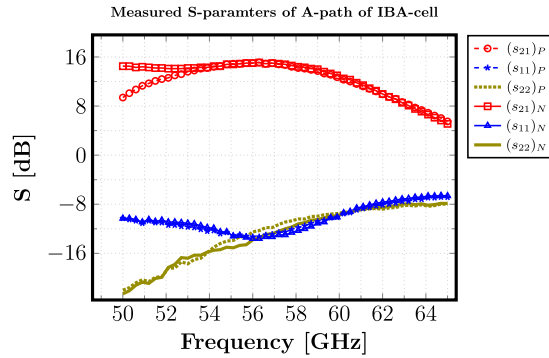
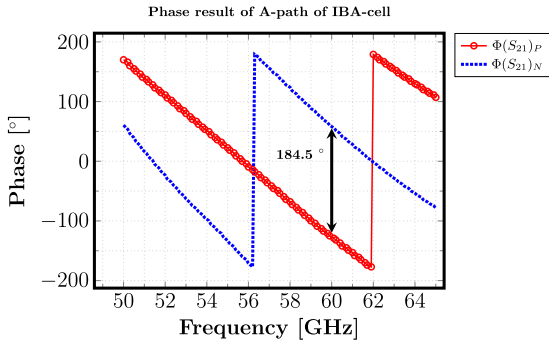


Fig. 21. Measured S-parameters versus simulated for the IBA-cell at the maximum gain state.



(a)

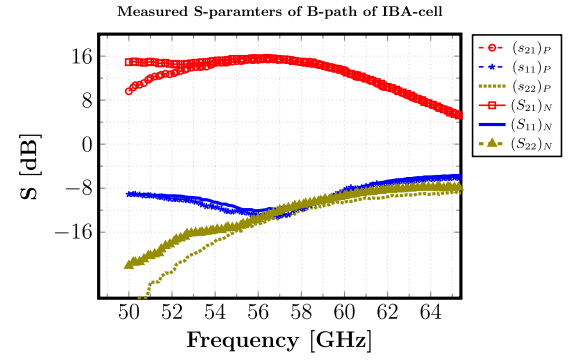


(b)

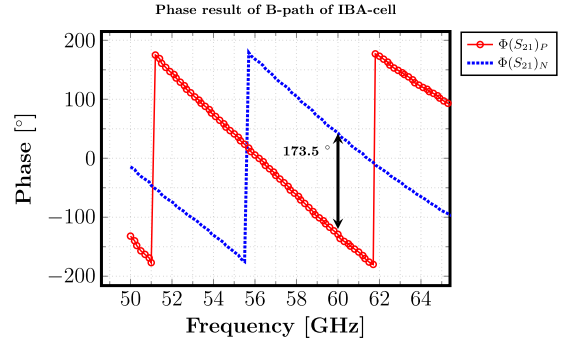
Fig. 22. Measured results of A-path IBA-cell for IBA front end. (a) S-parameters of the A-path of the IBA-cell. (b) Phase response of the A-path of the IBA-cell.

SW = “1.” The gain TR is from 15 to −2 dB. But, the phase variations in Fig. 28(b) are as large as 56°. The optimal phase compensation technique has been presented in the previous section, and the comparison of measured results is shown in Fig. 29: without phase compensation, the phase variation is over 20° at 60 GHz, plotted in the lines without markers, while the phase variation reduces to 10° with compensation.

Table V compares the measured results of the stand-alone test chips and the IBA-cell IC with the state-of-the-art works. The table summarizes the reported VGAs at mm-wave frequencies and lists the performance in terms of NF, gain TR, IIP₃, and power consumption. The design of the VGLNA validates the low IIP₃ variation and the simultaneous low NF



(a)



(b)

Fig. 23. Measured results of B-path IBA-cell for IBA front end. (a) S-parameters of the B-path of the IBA-cell. (b) Phase response of the B-path of the IBA-cell.

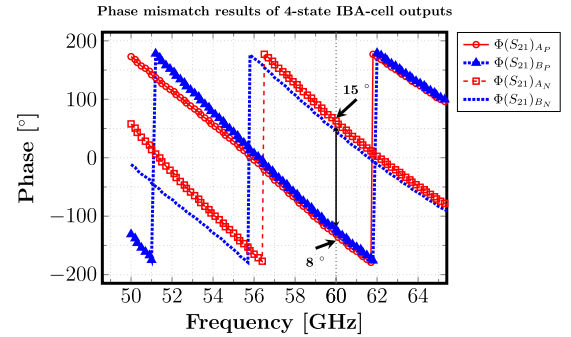


Fig. 24. Phase mismatch results of IBA-cell for IBA front end.

performance. Extending to the IBA-cell, this paper demonstrates the IIP₃ (variations) performance required for the IBA front-end receiver. The NF of the IBA-cell is higher compared to the stand-alone test chips and other reference works, trading off with cascading stages of high gain TR. The measured results establish the feasibility of the IBA cell. Future work will focus on the IBA cell integration with the succeeding blocks of phase shifters (reported in [19]) and VGA₂, for the full IBA beamforming architecture.

Normally, a figure of merit is used to compare the state-of-the-art amplifiers [38]

$$\text{FoM}_I = \frac{G[\text{abs}] \times \text{BW}[\text{GHz}]}{(F - 1) \times P_{\text{dc}}[\text{mW}]} \quad (3)$$

TABLE V
STATE-OF-THE-ART VGAS IN COMPARISON TO THE VGLNA AND IBA-CELL OF THIS PAPER

	This work IBA-cell	This work VGLNA	ASSCC2008 [36]	RFIC2016 [28]	MWCL2018 [40]	RFIC2016 [25]	ESSCIRC2016 [26]
Freq. (GHz)	50-61	56-65	57-64	48-61	26-33	26-30	15.5-39
Technology	40 nm CMOS	40 nm CMOS	65 nm CMOS	40 nm CMOS	40 nm CMOS	GF 8HP BiCMOS	BiCMOS
Arch.	4 Cascode 4 diff.	Cascode	Cascode	Cascode	1 Cas. 2 Diff.	2 CS	Cascode
NF(dB)	6.5	4.3*, 5.13 [†]	5.6	3.8	3.3-4.3	8.6 ^{††}	3.6
S21 (dB)	15.86 [‡]	6.67	15	12.5	27.1	18	17
Gain tuning range (dB)	-2~15	1.02~6.67	5~15	N/A	18~26	10~18	-6~17
IIP ₃ (dBm)	-11.3	-7.66	-6	-15	-6	-4 [‡]	-1
Area (μm^2)	1735 × 925	950 × 665	1400 × 750	650 × 310*	795 × 590	N/A	760 × 240*
Power (mW)	59.4 [‡]	12.1	46.5	20.4	21-31.4	N/A	104
FOM _I [*]	3.7 [◊]	1.6	1.6	7.2	96	N/A	8.9
FOM _I [†]	1.19 [◊]	1.9	0.28	N/A	0.65	N/A	155

[†] measured results of the VGLNA maximum gain state.

[‡] measured at 57 GHz.

[‡] includes DC power consumptions of two A/B-path VGAs.

* Chip core only.

◊ DC power including one path VGA.

* measured results of the LNA.

^{††} simulated result.

[‡] calculated from IP_{1dB} results, by $IIP_3 = IP_{1dB} + 9.6(\text{dB})$.

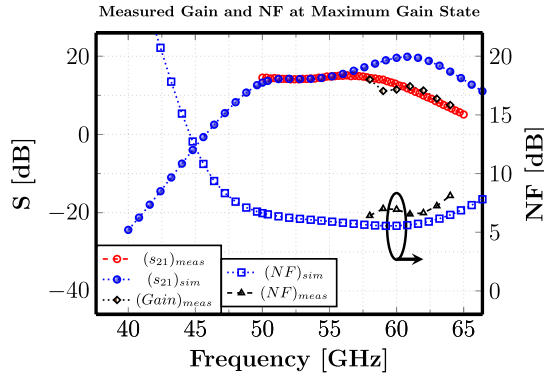


Fig. 25. NF and gain measured results for the NF measurement setup.

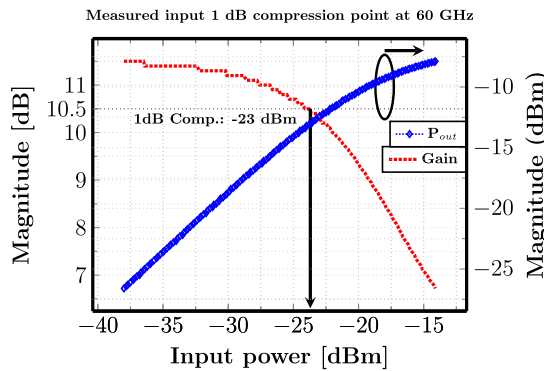


Fig. 26. Measured IP1dB at 60 GHz.

This does not include the characteristics of VGAs and IC implementation, for example, gain TR, IIP₃, and chip areas; therefore, an extended FoM_{II} is utilized in this paper [39]

$$\text{FoM}_{II} = \frac{\text{BW}[\text{GHz}] \times \text{IIP}_3[\text{mW}] \times \text{TR}[\text{abs}]}{(F - 1) \times P_{\text{dc}}[\text{mW}] \times \text{area}[\mu\text{m}^2]}. \quad (4)$$

As seen from Table V, [40] has the highest FoM_I due to the highest power gain at the center frequency of 28 GHz.

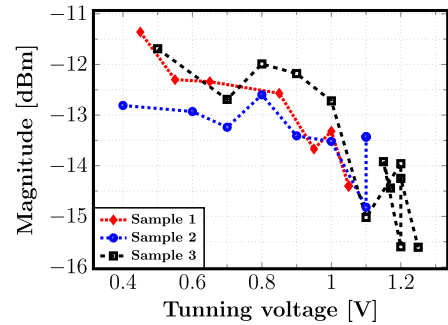


Fig. 27. Measured IIP₃ at 60 GHz.

Similarly, [28] has twice FoM_I compared to this paper thanks to state-of-the-art NF achieved; however, it does not have the ability of gain tuning, and its single-ended implementation need to extend as a differential structure for the whole array integration, which means the power consumption would increase and the NF would degrade. This paper achieves a considerable performance, trading off between the low NF requirement and the gain tuning ability. Reference [26] achieves low NF and high TR by the advantage of BiCMOS technology; however, the concern about it is that the BiCMOS technology is not as friendly as CMOS to implement advance array algorithms or calibration schemes.

5) *System Simulations*: According to the measured and optimized simulation results, the work has been done to back estimate the IBA system performance in terms of array pattern, and Fig. 30 is plotted via MATLAB, where the number of antenna channels is 9 with the interpolation factor of 4. In Fig. 30(a), the incident angle is at 135°, and a uniform linear weighing algorithm is applied; therefore, the IBA system requires about 6-dB gain TR for a considerable system performance [15]. There are three cases estimated.

- 1) *Sim.1*: TR = 9 dB, phase variation $\Delta = 1^\circ$.
- 2) *Meas.1*: TR = 17 dB, phase variation $\Delta = 56^\circ$.

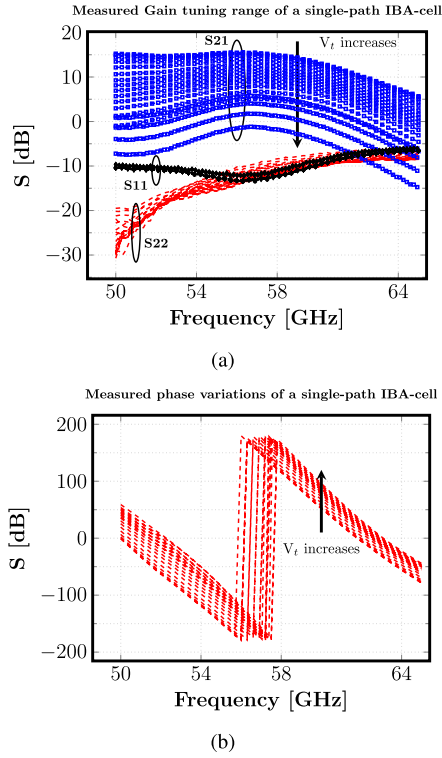


Fig. 28. Measured TR of the IBA-cell. (a) S-parameters. (b) Phase response.

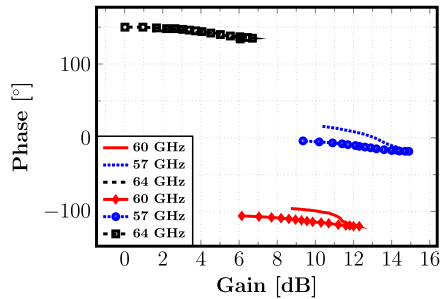


Fig. 29. Phase variations over the gain TR at different frequencies: 57, 60, and 64 GHz. Lines without markers: without phase compensation techniques. Lines with markers: with phase compensation techniques.

- 3) *Meas.2*: TR = 5.6 dB, phase variation $\Delta = 7^\circ$.
- 4) *Sim.2*: TR = 17 dB, phase variation $\Delta = 7^\circ$.

First of all, the array pattern of case 2 shows that the implemented IBA-cell performance is good enough for a uniform linear array (see Fig. 30). To further discuss the circuit performance effect on the array pattern, Fig. 30(a) shows that, for the peak of main lobe, compared with case 1, the phase variations and the gain TR cause the peak degradation while some sidelobes' amplitude level increases. It shows that the gain TR is the dominant factor for the sidelobe level, observed from the curves of cases 2 and 3. If the phase variation is reduced from 56° to 7° (cases 2 and 4), it improves the nulls particularly and the nulling is mainly determined by the factor of phase variation according to cases 3 and 4. The imperfection of the weighting VGA can lead to higher level of sidelobes and degradation of nulls at some directions; therefore,

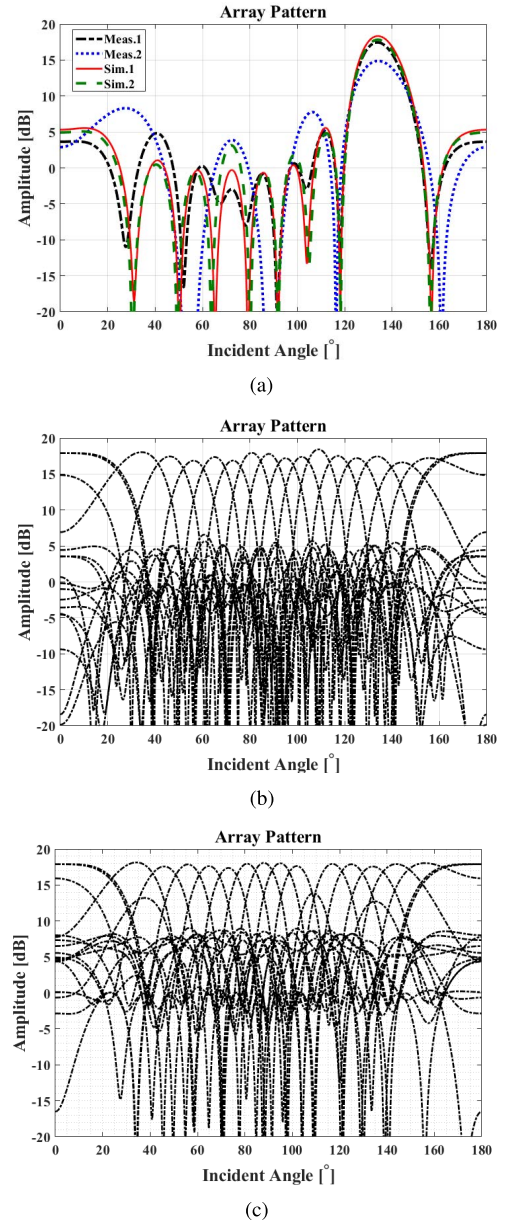


Fig. 30. Simulated array patterns using the simulated/measured VGLNA S-parameter data, of which the interpolation factor equals to 4, and the number of channels is 9 [15]. (a) Array pattern comparisons for: 1) Sim.1: TR = 9 dB, $\Delta = 1^\circ$; 2) Meas.1: TR = 17 dB, $\Delta = 56^\circ$; 3) Meas.2: TR = 5.6 dB, $\Delta = 7^\circ$; and 4) Sim.2: TR = 17 dB, $\Delta = 7^\circ$. (b) Array pattern of the full 180° steering range, where Meas.1 data are applied. (c) Array pattern of the full 180° steering range, where Meas.2 data are applied.

the main-lobe-to-sidelobe ratio reduces to 10 dB for case 3, while it remains about 13 dB when the TR of 17 dB applied in case 2. Comparing between cases 2 and 3, the influence of gain TR and phase variation is not a linear superposition. It suggests that a successful phase variation compensation technique can relax the TR of the weighting amplifier; therefore, it hopes to save power and cost. Furthermore, for the full 180° steering range, the array patterns of cases 2 and 3 are plotted in Fig. 30(b) and (c), respectively. Besides the main-lobe-to-sidelobe ratio degradation in case 3, it is observed that there are the dips at some specific directions due to the limited TR of 5.6 dB for weighting. It agrees with the work in [15].

V. CONCLUSION

In this paper, the implementation of the key design blocks for the IBA beamforming architecture [15] is presented. The importance for the implementation of the IBA cell, in terms of low NF, moderate linearity, and small linearity deviation for a high gain TR, is demonstrated.

The design techniques for a VGLNA have been explained and verified with a single-ended LNA and VGLNA chip for the 60-GHz band. Then, the full integrated differential IBA-cell is presented and measured. The results indicate that the integration of the IBA-cell has been demonstrated according to the specifications derived in [15]. The bandwidth and NF are achieved while the TR of the weighting amplifier extends from 6 to 17 dB; the best IIP₃ degrades from −7 to −11 dBm, with the deviation from −11.3 to −16 dBm, which is expected for a 4-stage (or 6-stage cascode) amplifier and meets the required specs. The phase response variations still need further improvement in order to achieve the overall beamforming performance by the proposed phase compensation technique. The dc power consumption is 59 mW. The system simulation indicates that the implementation of the IBA cell in this paper has successfully met the system requirements and ready for further integration. Also, it shows the insight of the design of the IBA-cell for integration: for circuit level, the IBA-cell can reduce the gain TR to save power consumption, by means of applying the proposed phase compensation technique of cross-coupled feedback loop, and meanwhile, the proposed phase compensation technique has the advantage of reducing the linearity degradation at low gain states; for the system level, the higher the phase variations in the IBA-cell circuits over all the gain states, the higher IIP₃ required by the circuits from the perspective of in-band performance, because the level of nulling is reduced. Overall, this paper validates the idea and the circuit implementation of the novel beamforming architecture presented in [15]. Future work will focus on the integration with the phase shifter [19].

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